

Design and Development of a Framework for Flight Dynamics and Control for Small Conventional Surveillance UAVs

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Introduction & Background

Small fixed-wing UAVs are rapidly expanding across surveillance, agriculture, disaster response, and reconnaissance. Their effectiveness critically depends on flight dynamic behavior and flying qualities, which are fundamentally governed by the geometric configuration, mass distribution, and inertial properties of the airframe. Current UAV development relies heavily on iterative prototyping and flying qualities deficiencies are identified only after fabrication and corrected through autopilot tuning rather than deliberate geometric design. A systematic methodology that directly links UAV geometry to dynamic behavior is absent in existing literature, particularly for small conventional surveillance UAVs.

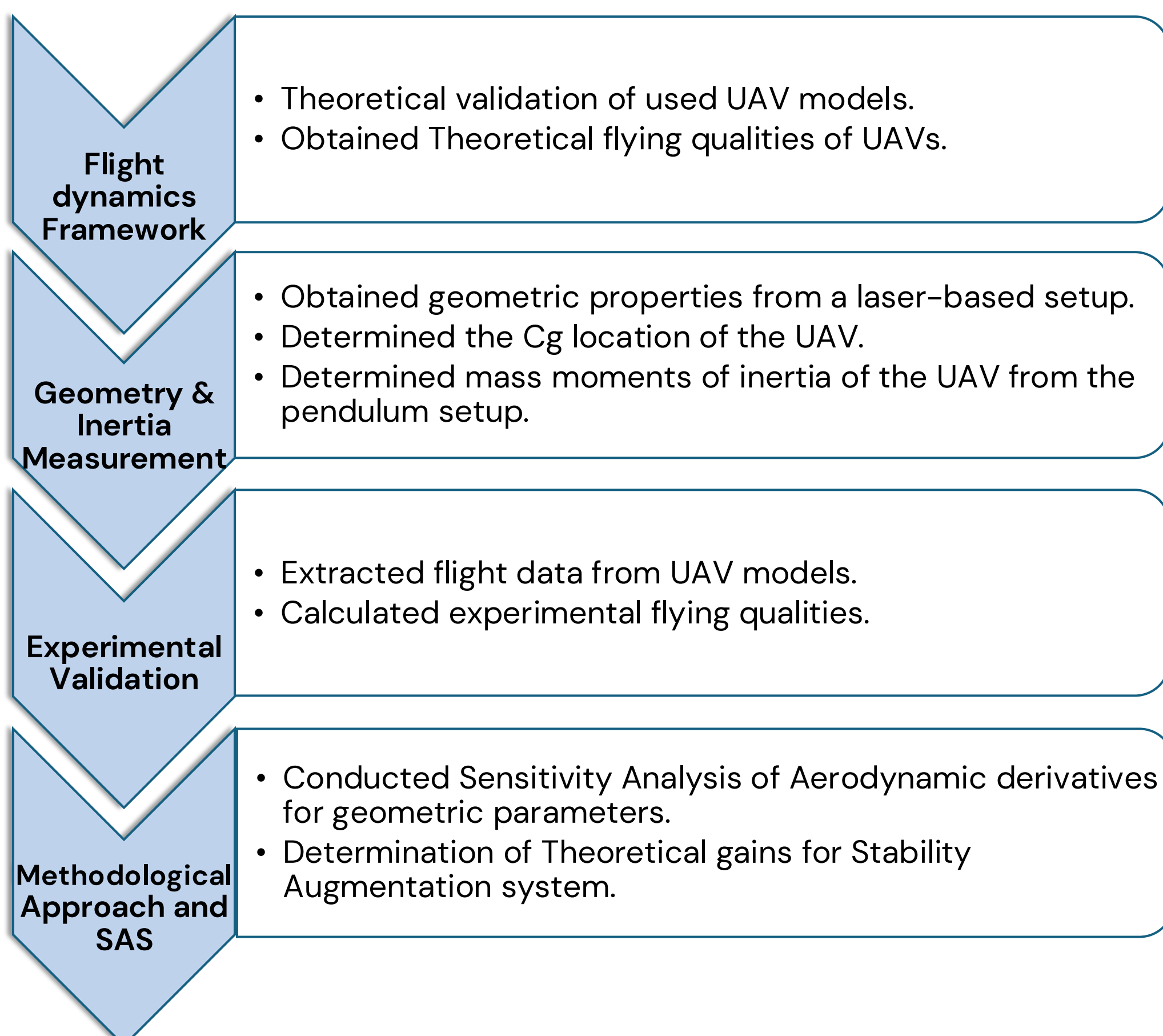
Aim & Objectives

Aim : To design and develop a framework for small conventional surveillance UAVs that enables reliable analysis of flight dynamics, stability, and control behavior.

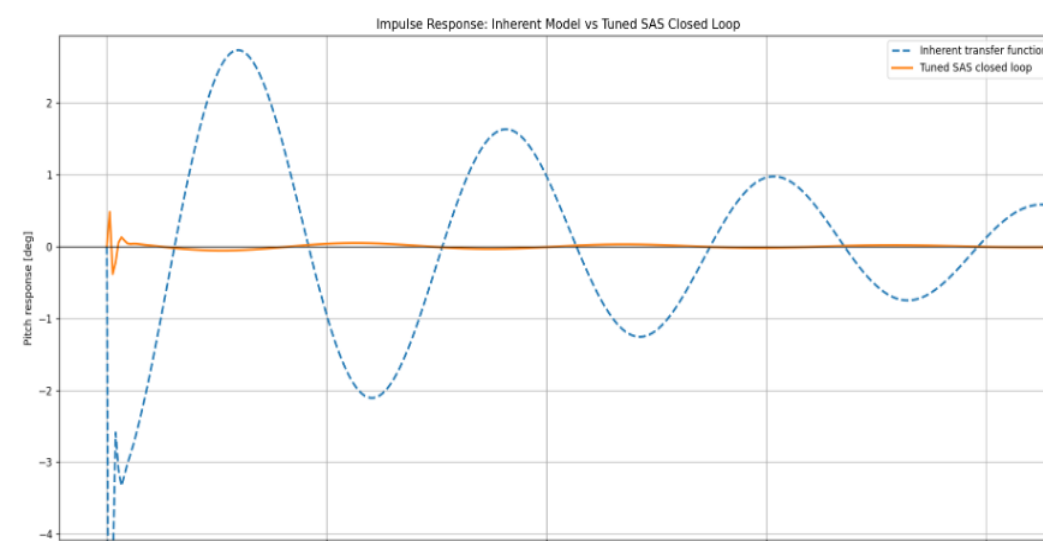
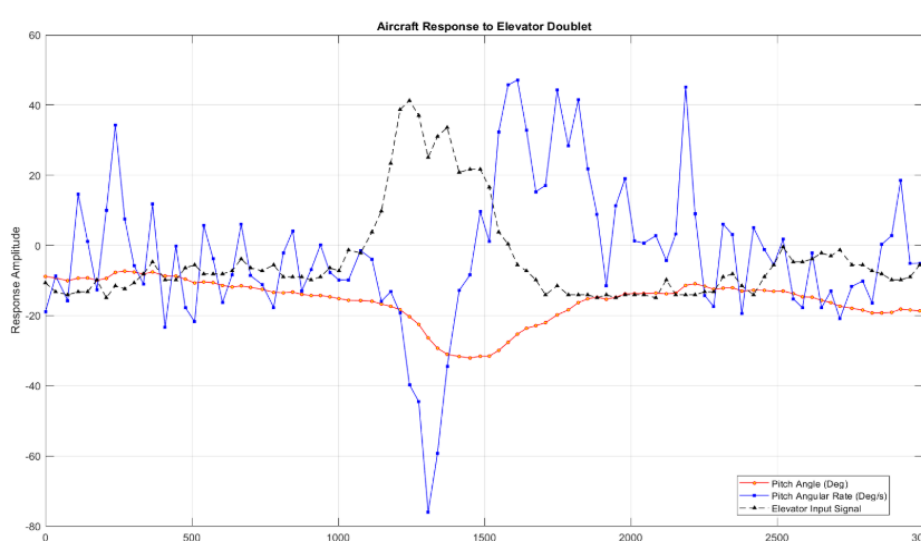
Objectives

1. Develop the Flight Dynamic Framework using governing principles.
2. Fabrication of the experimental setup to measure geometric and inertia properties of a small UAV.
3. Experimentally validating the Flight Dynamic model for selected modes.
4. Propose a methodological approach to achieve the expected flying qualities for surveillance UAVs and develop a stability augmentation system.

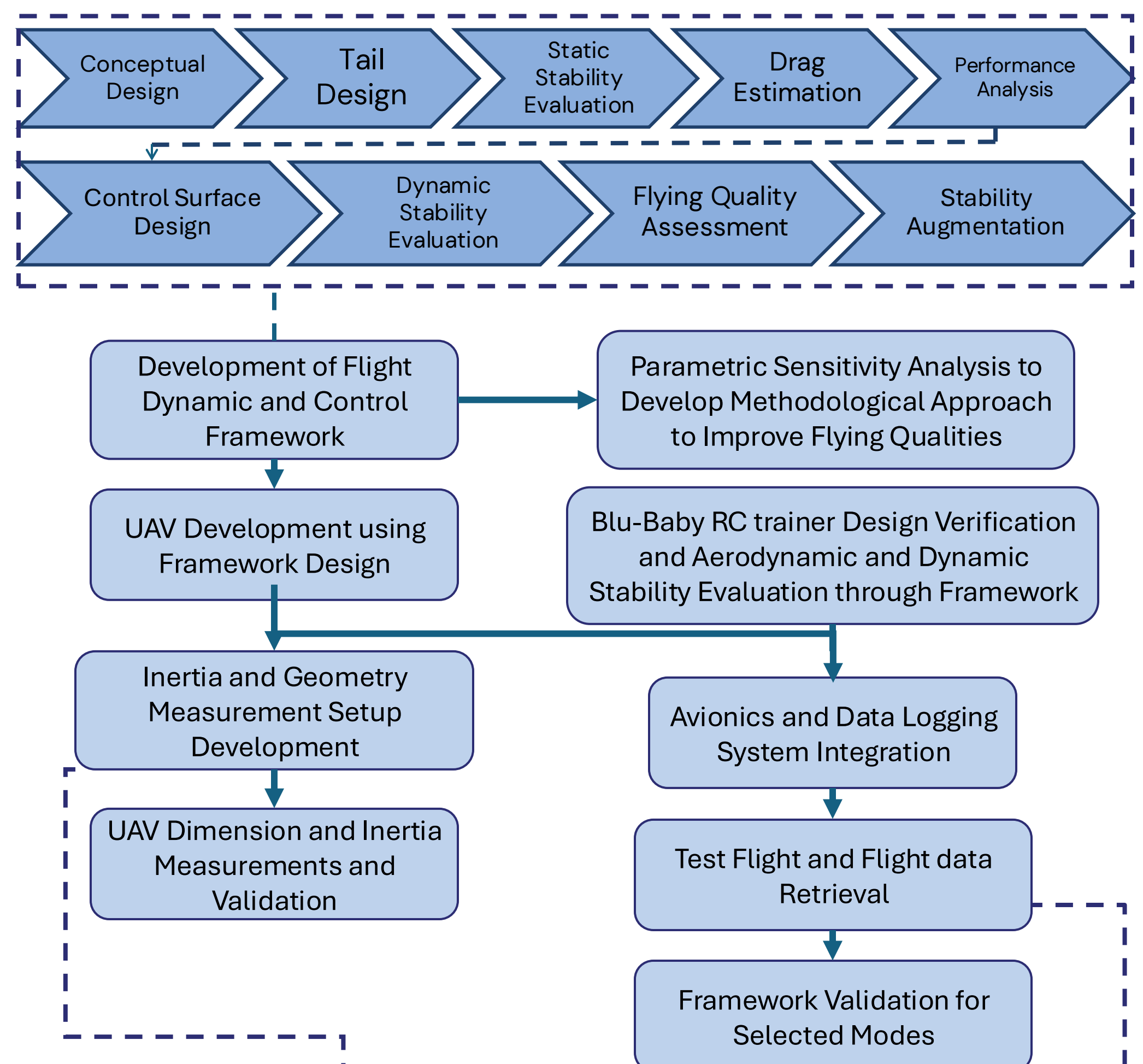
Results & Conclusion



Conclusion: A unified framework was developed to evaluate UAV dynamics, and stability. Experimental measurements and flight tests supported model validation, while sensitivity analysis identified key design parameters for improving flight stability and handling qualities.



Methodology



Future Works

- ❖ Use sensitivity data from multiple UAV configurations to train a machine learning model that identifies patterns between geometry and dynamic behavior for efficient flight dynamics tuning.
- ❖ Improving the laser measurement system accuracy by introducing vernier scales.
- ❖ Improving Roll/Yaw SAS through state space model methods.

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